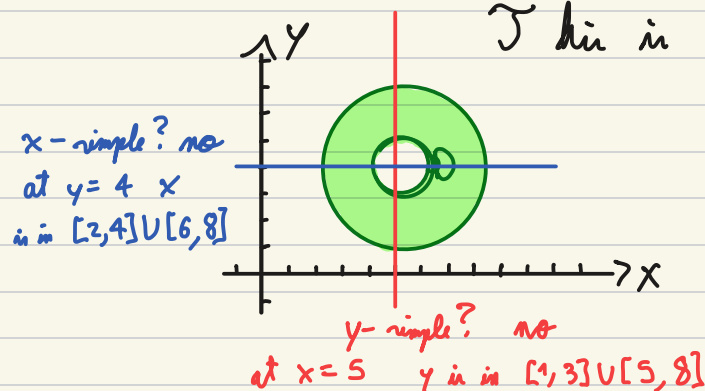


April 3, 2026

• Example?

This is not an elementary region



Integrating over simple regions

1) **y-simple**, if we can describe it as

$$D = \{(x, y) \in \mathbb{R} : a \leq x \leq b, \gamma(x) \leq y \leq \delta(x)\}$$

for some continuous functions $\gamma, \delta: [a, b] \rightarrow \mathbb{R}$

$$\int_a^b \int_{\gamma(x)}^{\delta(x)} f \, dy \, dx$$

2) **x-simple**, if we can describe it as

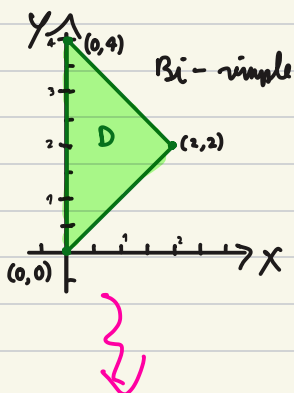
$$D = \{(x, y) \in \mathbb{R} : a \leq y \leq b, \alpha(y) \leq x \leq \beta(y)\}$$

for some continuous functions $\alpha, \beta: [a, b] \rightarrow \mathbb{R}$

$$\int_a^b \int_{\alpha(y)}^{\beta(y)} f \, dx \, dy$$

e.g. integrate $f(x,y) = x^2 y$

over the triangle with vertices $(0,0)$, $(2,2)$, $(0,4)$



$$D = \{(x,y) \in \mathbb{R}^2 : 0 \leq x \leq 2, x \leq y \leq -x+4\}$$

$$\int_0^2 \int_x^{4-x} x^2 y \, dy \, dx$$

or y -simple region $D = \{(x,y) \in \mathbb{R}^2 : 0 \leq x \leq 2, x \leq y \leq 4-x\}$

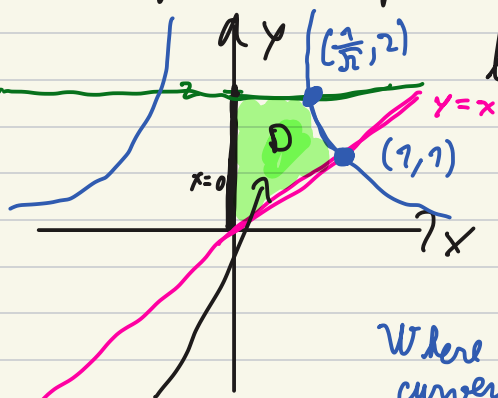
or x -simple region $D = \{(x,y) \in \mathbb{R}^2 : 0 \leq y \leq 4, 0 \leq x \leq 2-|y-2|\}$

note: not all simple descriptions are equally simple

$$\begin{aligned} & \int_0^2 \left[\int_x^{4-x} x^2 y \, dy \right] dx \\ &= \int_0^2 \left[x^2 \int_x^{4-x} y \, dy \right] dx = \int_0^2 x^2 \left[\int_x^{4-x} y \, dy \right] dx \\ &= \int_0^2 x^2 \left[\frac{y^2}{2} \Big|_{y=x}^{y=4-x} \right] dx = \int_0^2 x^2 \left[\frac{(4-x)^2}{2} - \frac{x^2}{2} \right] dx \\ &= \int_0^2 x^2 \left[8 - 4x + \frac{x^2}{2} - \frac{x^2}{2} \right] dx \\ &= \int_0^2 8x^2 - 4x^3 \, dx = \left(\frac{8}{3} x^3 - x^4 \right) \Big|_{x=0}^{x=2} \\ &= \frac{64}{3} - 16 = \frac{64-48}{3} = \frac{16}{3} \end{aligned}$$

Non elementary region

e.g. integrate $3y$ over the region



bounded by:

$$y=2, \quad y=x, \quad x=0, \\ x^2 y \leq 1 \rightarrow y = \frac{1}{x^2}$$

Where does $y = \frac{1}{x^2}$ intersect the curve that we have above:

$x \neq 0$: no intersection with $x=0$
intersecting $y=2$: $2 = \frac{1}{x^2} \quad x = \pm \frac{1}{\sqrt{2}}$

$\Downarrow$

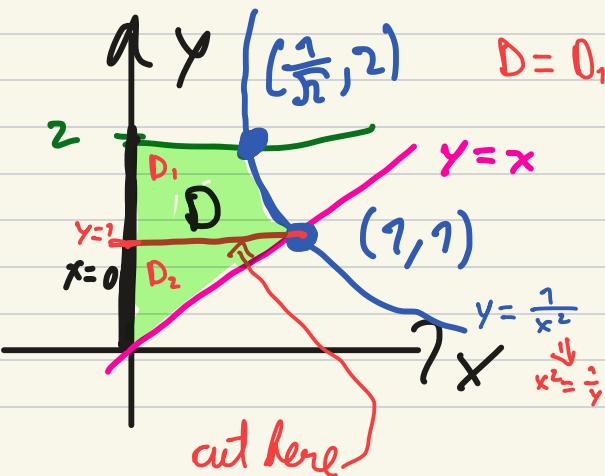
$$x = \frac{1}{\sqrt{2}}$$

intersecting $y=x$: $y=x = y = \frac{1}{x^2} \quad 1 = x^3 \rightarrow x=1$

$$y=x=1$$

This is the region bounded by those curves

D is bi-simple, but the description as x -simple or y -simple is not very easy, so we split the region

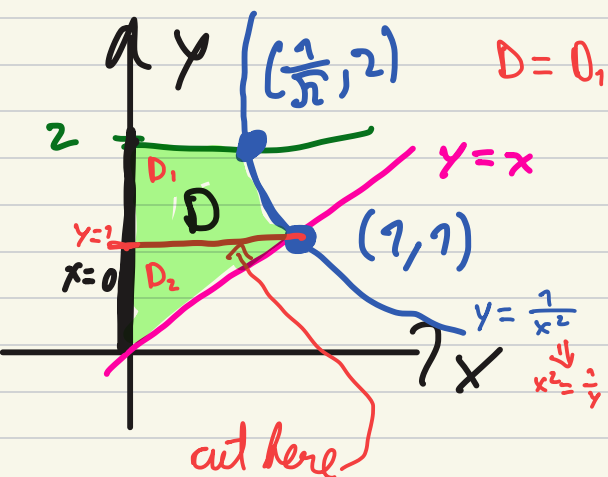


$$D = D_1 \cup D_2$$

$$D_1 = \{(x,y) \in \mathbb{R}^2 : 1 \leq y \leq 2, 0 \leq x \leq \frac{1}{\sqrt{y}}\}$$

$$D_2 = \{(x,y) \in \mathbb{R}^2 : 0 \leq y \leq 1, 0 \leq x \leq y\}$$

$$x^2 = \frac{1}{y} \rightarrow x = \frac{1}{\sqrt{y}}$$



$$D = D_1 \cup D_2$$

$$D_1 = \{(x,y) \in \mathbb{R}^2 : 1 \leq y \leq 2, 0 \leq x \leq \frac{1}{\sqrt{y}}\}$$

$$D_2 = \{(x,y) \in \mathbb{R}^2 : 0 \leq y \leq 1, 0 \leq x \leq y\}$$

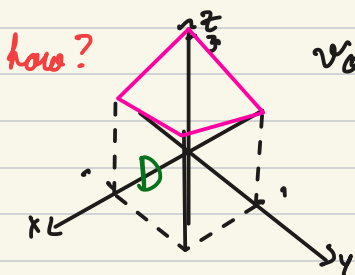
$$\begin{aligned} \iint_D 3y \, dA &= \iint_{D_1} 3y \, dA + \iint_{D_2} 3y \, dA \\ &= \int_1^2 \left[\int_0^{\frac{1}{\sqrt{y}}} 3y \, dx \right] dy + \\ &\quad \int_0^1 \left[\int_0^y 3y \, dx \right] dy \end{aligned}$$

From here on is just a matter of integrating and being careful about which variable we are integrating with respect to

Triple integrals

motivation: Let's find the volume of the region in $\mathbb{R}^3$ bounded by the surfaces:

$$z=0, x=0, y=0, x=1, y=1, z=3-x-y$$



Volume under $z = 3 - x - y$ over the rectangle $[0, 1] \times [0, 1] = R$

as an iterated double integral

$$V = \iint_R (3 - x - y) \, dA$$

$$= \int_0^1 \int_0^1 (3 - x - y) \, dx \, dy$$

or a triple integral $= \int_0^1 \int_0^1 \left[\int_0^{3-x-y} 1 \, dz \right] dx \, dy$

$$= \int_0^1 \int_0^1 \left(z \Big|_{z=0}^{z=3-x-y} \right) dx \, dy$$

clf. we want to integrate any continuous function f on D (the solid region in $\mathbb{R}^3$ described above)

we do $\int_0^1 \int_0^1 \int_0^{3-x-y} f(x, y, z) \, dz \, dx \, dy$

e.g. What is the volume

$$D = \{ (x, y, z) \in \mathbb{R}^3 : 0 \leq x \leq 1, 0 \leq y \leq 1 - x, 0 \leq z \leq 1 - y - x \}$$

